

Emily Biondi, MSN, APRN-CNM
Certified Nurse Midwife and Biomedical Engineer
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EDUCATION

Case Western Reserve University <i>PhD, Biomedical Engineering</i>	August 2026-Present <i>Cleveland, OH</i>
Case Western Reserve University, Frances Payne Bolton School of Nursing <i>MSN, Nurse Midwifery (GPA 4.0/4.0)</i>	May, 2023 <i>Cleveland, OH</i>
Case Western Reserve University, Frances Payne Bolton School of Nursing <i>MN, Graduate Entry (GPA 4.0/4.0)</i>	May, 2021 <i>Cleveland, OH</i>
University of Virginia, School of Engineering and Applied Science <i>BS, Biomedical Engineering (GPA 3.9/4.0)</i>	May, 2017 <i>Charlottesville, VA</i>

EXPERIENCE

Louis Stokes Cleveland VA Medical Center and CWRU <i>Graduate Researcher and Research Nurse</i>	May 2026 – Present <i>Cleveland, OH</i>
University Hospitals <i>Certified Nurse Midwife</i> - Provided comprehensive gynecological, prenatal, intrapartum, and postpartum care to families in a large academic medical center - Provided individualized and culturally sensitive gynecologic care and family planning services	Oct 2025 – Dec 2025 <i>Cleveland, OH</i>
Mt Eaton Midwifery <i>Certified Nurse Midwife</i> - Delivered comprehensive antepartum, intrapartum, and postpartum care to Amish and Mennonite patients at a free-standing, religious-exempt birth center - Worked as part a team of 4 CNMs to manage approx. 15-20 births monthly, promoting low-intervention and evidence-based care in a community setting	Jul 2023 – Sept 2025 <i>Dundee, OH</i>
Case Western Reserve University Health & Counseling Services <i>Registered Nurse</i> - Triaged patients and conducted point-of-care testing in high-volume acute and respiratory care clinics at a university serving over 12,000 students, ensuring timely and appropriate clinical interventions - Collaborated with a multidisciplinary team of 10 RNs, 5 APRNs and 4 MDs to provide efficient and accessible student-centered healthcare - Educated patients on health management strategies, including management of acute illnesses as well as preventative care to support overall wellness	Sep 2021 – Dec 2022 <i>Cleveland, OH</i>
Louis Stokes Cleveland VA Medical Center and CWRU <i>Research Biomedical Engineer/ Graduate Researcher</i> - Assisted in clinical trials investigating the effects therapeutic neuromodulation on restoring bladder function after neural disease or trauma, such as spinal cord injury - Responsible for development and maintenance of equipment and systems, data collection, data processing, and data analysis, and contributed to interpretation of results - Developed, organized, and submitted clinical research documentation including study protocols, informed consent forms, and data management plans in coordination with the Institutional Review Board	Aug 2017 – Jul 2019 <i>Cleveland, OH</i>

Department of Biomedical Engineering, University of Virginia**Oct 2015 – July 2017***Undergraduate Research Assistant**Charlottesville, VA*

- Assisted with device development and design to prevent growth of antibiotic-resistant bacterial biofilms in hospital plumbing systems

Department of Radiology and Medical Imaging, University of Virginia**May 2014-August 2015***Undergraduate Research Assistant**Charlottesville, VA*

- Evaluated the Bayer Radimetrics Enterprise Platform, a dose management and analytics software for computed tomography (CT), to assess its effectiveness and integration.
- Assisted in hyperpolarized helium and xenon magnetic resonance imaging (MRI) studies focused on pulmonary imaging and gas exchange, conducting research subject interviews and supporting clinical study procedures

Department of Biomedical Engineering, University of Virginia**Spring-Summer 2015***Undergraduate Research Assistant**Charlottesville, VA*

- Assisted with research on the modulation of neovascularization in post-myocardial infarct zones using hornerin-targeted liposomes

TEACHING

- Clinical Preceptor** – Case Western Reserve University, MSN Program
 - Preceptor to Abby Henkel, RN, BSN (Mar 2025- May 2025)
- Graduate Mentor** – Case Western Reserve University, Department of Biomedical Engineering
 - Advised and supported undergraduate research students in the Gustafson lab at Case Western Reserve University

AWARDS

- Awarded Dean's Legacy Award for performance in MSN program
- Awarded MN Alumni Award for Outstanding Clinical Competence in MN program

VOLUNTEER EXPERIENCE

- Sep 2021- May 2022: Student Representative, Ohio Affiliate of ACNM
- May 2020 – Aug 2020 Nurse Intern, University Hospitals – Lerner Tower 3
- Sep 2019 – Feb 2020: Volunteer, CWRU Student-run Health Clinic
- Jun 2018-Jul 2019 Volunteer, MetroHealth Emergency Department

CERTIFICATIONS

- Certifications:** Registered Nurse (RN), Certified Nurse Midwife (CNM), Neonatal Resuscitation Program (NRP), Basic Life Support (BLS)

PUBLICATIONS & PRESENTATIONS

Abstracts

- Mugler, E.A.,** K.N. Miller, A.R. Goode, P.G. Judy, T.A. Altes, P. Komlosi, and S. Mukherjee (2015) "Retrospective Review of Head CT Doses During a 6 Month Period: A Method for Identifying Opportunities for Improvement through the Prevention of Outliers" *American Society of Neuroradiology 53rd Annual Meeting*, April 25-30, 2015.
- Dasa, S.S.K., R. Suzuki, **E. Mugler**, L. Chen, R. Jansson-Löfmark, E. Michaëlsson, L. Lindfors, A.L. Klivanov, B.A. French and K. A. Kelly (2016) "Neovascularization in the infarct zone post-MI can be modulated using hornerin-targeted liposomes for endothelial-specific delivery of VEGFR2 inhibitor" *World Molecular Imaging Congress*, 2016.
- Chen, W., P. Wilson, V. Sama, H. Stecko, J. Volmer, **E. Mugler**, S. Meade, D. Bourbeau, S. Brose, J. Dunning, J.

Schnellenberger, K. Gustafson (2018). “Hardware and Software Validation of a Novel Wireless Surface Stimulator.” CWRU SOURCE Intersections.

Papers

- Gutknecht, M.F., Seaman, M.E., Ning, B., Cornejo, D.A., **Mugler, E.**, Antkowiak, P.F., Moskaluk, C.A., Hu, S., Epstein, F.H., and Kelly, K.A. (2017). Identification of the S100 fused-type protein hornerin as a regulator of tumor vascularity. *Nature Communications* 8, 552.

Posters Presented

- Bourbeau, D., C. Nguyen, **E. Mugler**, R. Karam, S. Majerus, J. Wilson, G. Nemunaitis, K. Gustafson, and M. Damaser (2017) “Automated closed-loop stimulation to inhibit neurogenic bladder overactivity.” MetroHealth Medical Center Research Days. October 24, 2017. Cleveland, OH.
- Bourbeau, D., S. Brose, **E. Mugler**, M. Richmond, and K. Gustafson (2018) “Reducing Bladder Reflexes after Spinal Cord Injury: ‘Bladder Pacemaker.’” VA Research Week 2018. May 17, 2018. Cleveland, OH.
- **Mugler, E.** (2022). “Pelvic Floor Therapy and Sexual Dysfunction”. Digital Poster Presentation: Ohio ACNM Forward, Newark, OH February 5, 2022